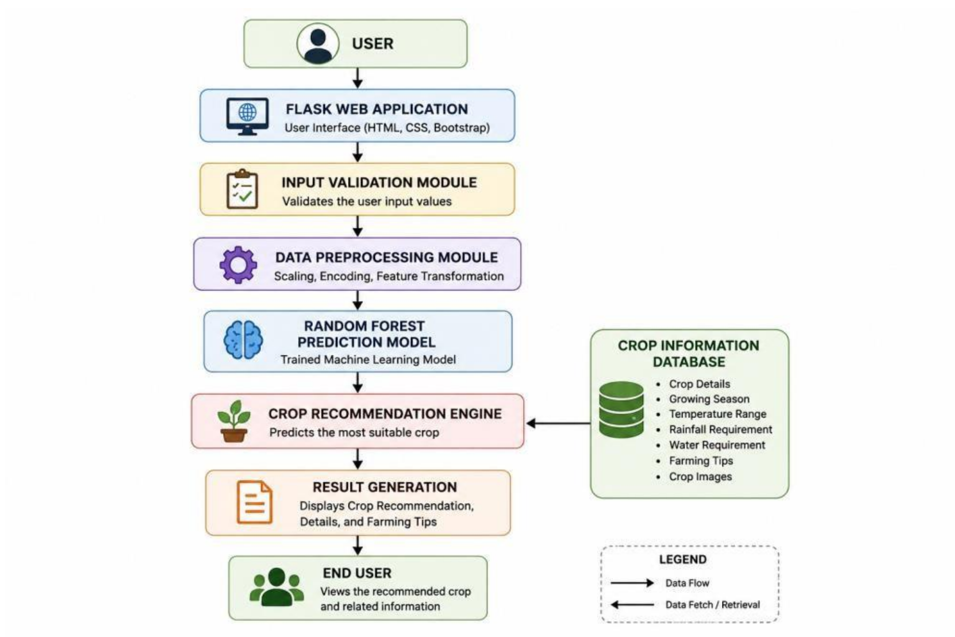
Solution Architecture

Date	14 July 2026
Team ID	
Project Name	OptiCrop – Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Solution Architecture Diagram:



Component Description Table:

Component Name	Description / Role	Technologies Used
Presentation Layer	Collects soil and climate details from the farmer and displays the crop recommendation.	Flask, HTML, CSS, Bootstrap
API / Request Handler	Routes prediction requests from the interface to the ML service.	Flask
Core Logic Service	Preprocesses input data and predicts the most suitable crop.	Python, Scikit-learn (Random Forest)
Crop Information Store	Stores the dataset, trained model, and crop information used to generate the final recommendation.	CSV, Joblib (.pkl)